

Ethics, Privacy, Security, Bias, and Responsible Data Use

DAEN 459 - Fall 2026
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Questions we want to address:

- What should this system refuse to do?
- What data should never be collected?
- What assumptions could fail in production?
- Who audits the system?
- How would users challenge an incorrect result?

Case study:

<https://www.youtube.com/watch?v=gnjVqeBoaqs>

Source map: governance and ethics

- **NIST AI RMF:** AI risk, trustworthiness, lifecycle governance
<https://www.nist.gov/itl/ai-risk-management-framework>
- **NIST Privacy Framework:** privacy risk management
<https://www.nist.gov/privacy-framework>
- **ACM Code of Ethics:** professional duties for computing work
<https://www.acm.org/code-of-ethics>
- **OECD AI Principles:** trustworthy AI and human-centered values
<https://oecd.ai/en/ai-principles>

Source map: security, law, and data stewardship

- **OWASP Top 10:** common web application security risks
<https://owasp.org/www-project-top-ten/>
- **GDPR Article 5:** lawfulness, fairness, transparency, minimization
<https://eur-lex.europa.eu/eli/reg/2016/679/oj>
- **FAIR Principles:** findable, accessible, interoperable, reusable data
<https://www.go-fair.org/fair-principles/>

Source map: documentation practices

- **Datasheets for Datasets:** document dataset motivation, composition, collection, preprocessing, uses, and risks
<https://arxiv.org/abs/1803.09010>
- **Model Cards:** document model purpose, evaluation, intended use, limitations, and ethical considerations
<https://arxiv.org/abs/1810.03993>

How to use these sources in capstone projects

- Use **NIST AI RMF** to identify and govern AI-related risks.
- Use **NIST Privacy Framework** and **GDPR principles** to justify privacy controls.
- Use **OWASP Top 10** to evaluate application and API security.
- Use **FAIR principles** to improve data stewardship.
- Use **Datasheets** and **Model Cards** as required documentation templates.

Why responsible data engineering matters

- Data pipelines create real-world consequences.
- Ethical failures often begin as design shortcuts.
- Responsible systems require technical, legal, and social controls.

Core responsibilities

- Collect only what is needed.
- Protect data across its lifecycle.
- Document provenance, consent, assumptions, and limitations.
- Test for bias, leakage, misuse, and downstream harm.
- Make systems auditable and governable.

Privacy

- Data minimization
- Purpose limitation
- Consent and notice
- De-identification and re-identification risk
- Retention and deletion policies

Security

- Authentication and authorization
- Encryption in transit and at rest
- Secrets management
- Least privilege
- Logging, monitoring, and incident response

Bias and fairness

- Historical bias
- Sampling bias
- Label bias
- Measurement bias
- Deployment feedback loops

Responsible data lifecycle

- (1) Define purpose and stakeholders.
- (2) Assess data sources and permissions.
- (3) Build secure pipelines.
- (4) Validate quality and representativeness.
- (5) Document assumptions and risks.
- (6) Monitor performance and harms after deployment.

Project risk checklist

- Who could be harmed by errors?
- What sensitive attributes are present or inferable?
- Could the data be re-identified?
- Who has access to raw data?
- Are outputs explainable and contestable?
- What happens when the model or pipeline fails?

Documentation artifacts

- Data dictionary
- Data lineage diagram
- Dataset datasheet
- Security threat model
- Bias and fairness assessment
- Retention and deletion policy
- Model card, if ML is used

Capstone deliverables

- Architecture diagram
- Data governance plan
- Privacy and security controls
- Bias-risk analysis
- Responsible-use statement
- Monitoring and rollback plan

Team Homework: Responsible Data Use Review

Goal for next Friday lab: prepare a 3–5 minute team presentation on your project's ethics, privacy, security, bias, and responsible data-use risks.

Analyze your project

- What data do you collect, store, or process?
- What privacy or security risks exist?
- What bias or fairness concerns could arise?
- Who could be harmed by incorrect results?

Present your plan

- Project context and data used
- 2–4 top responsible-use risks
- Concrete design or governance responses
- One open question for class feedback

Use at least two references: NIST AI RMF, NIST Privacy Framework, ACM Code of Ethics, OWASP Top 10, OECD AI Principles, GDPR, FAIR, Datasheets for Datasets, or Model Cards.

Guiding question: What could go wrong with our project, and what are we doing about it?